
Group Relative Policy Optimization for Control

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Abstract

This report studies Group Relative Policy Optimization (GRPO) for control. The full report will present a GRPO variant for continuous and discrete control tasks and evaluate it against PPO, SAC, and TD3 on Car Racing, cartpole-swingup-v0, and acrobot-swingup-v0. The midway version focuses on the required interim components: related work, MDP notation, and baseline results for PPO, SAC, and TD3.

1 Introduction

2 Related Work

3 Methodology

4 Theory

5 Experiments

6 Conclusion

References

A Proofs